

Wazuh + Suricata IDS Integration

Network Intrusion Detection & Auto-Block

Complete Step-by-Step Setup Guide

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Overview

This guide covers the complete setup of Suricata IDS integrated with Wazuh SIEM to automatically detect network attacks and block malicious IPs. Written from a real working environment — every command was tested and confirmed working.

What this setup does:

- Suricata monitors all network traffic in real-time on the agent machine
- When a suspicious packet matches a rule (50,861 rules loaded), Suricata writes an alert to eve.json
- Wazuh Agent reads eve.json and forwards alerts to Wazuh Manager
- Custom rule 86701 fires at Level 7 for any Suricata alert
- Active Response runs suricata-drop.sh — the attacker's IP gets blocked via iptables automatically
- Alert is visible in Wazuh Dashboard under Security Events

Environment used:

Component	Details
Wazuh Manager	Ubuntu Server 22.04 (hostname: wazuh-soc, IP: 103.228.38.219)
Wazuh Agent	Ubuntu Desktop 22.04 (hostname: rakib-virtual-machine, IP: 192.168.79.131)
Wazuh Version	4.14.5
Suricata Version	8.0.5
Network Interface	ens33

Step 1 — Install Suricata on the Agent

NOTE: Run all commands in this step on: AGENT (rakib-virtual-machine)

Add the official Suricata stable repository:

```
sudo add-apt-repository ppa:oisf/suricata-stable -y
sudo apt update
```

If suricata-update package conflict appears, remove it first:

```
sudo apt remove suricata-update -y
```

WARNING: Suricata 8.x includes suricata-update built-in. The old standalone package conflicts during upgrade. Always remove it before installing 8.x.

Install Suricata:

```
sudo apt install suricata -y
# Verify version
suricata --version
# Expected: This is Suricata version 8.0.5 RELEASE
```

Find your network interface:

```
ip a
# Look for the interface with your IP – in this setup: ens33
```

Step 2 — Configure Suricata

NOTE: Run on: AGENT (rakib-virtual-machine)

Download detection rules (Emerging Threats ruleset):

```
sudo suricata-update
# This downloads 50,000+ attack detection rules
# Rules saved to: /var/lib/suricata/rules/
```

Edit the main Suricata config file:

```
sudo nano /etc/suricata/suricata.yaml
```

Set your network interface (search with Ctrl+W for "af-packet"):

```
af-packet:
  - interface: ens33    # replace with your interface name

pcap:
  - interface: ens33    # same here
```

Ensure eve.json output is enabled (search for "eve-log"):

```
- eve-log:
  enabled: yes
  filetype: regular
  filename: eve.json
  types:
    - alert
    - http
    - dns
    - tls
    - flow
```

Validate config and start Suricata:

```
# Test config – look for "Configuration provided was successfully loaded"
sudo suricata -T -c /etc/suricata/suricata.yaml -v

# Enable and start
sudo systemctl enable suricata
sudo systemctl start suricata
sudo systemctl status suricata
# Expected: Active: active (running)
```

Verify eve.json is being written:

```
sudo tail -f /var/log/suricata/eve.json
```

```
# You should see JSON lines with event_type: stats, dns, etc.
```

```
# Press Ctrl+C to exit
```

Step 3 — Fix Permissions for Wazuh Agent

NOTE: Run on: AGENT (rakib-virtual-machine)

By default, /var/log/suricata/ is owned by the suricata group and is not readable by the wazuh user. This must be fixed or Wazuh will silently fail to read eve.json.

Check current permissions:

```
ls -la /var/log/suricata/  
# You will see: drwxrwx--- suricata suricata  
# The wazuh user cannot enter this directory
```

Add wazuh user to the suricata group:

```
sudo usermod -aG suricata wazuh
```

Restart Wazuh agent to apply group change:

```
sudo systemctl restart wazuh-agent
```

Verify wazuh can now read eve.json:

```
sudo -u wazuh cat /var/log/suricata/eve.json | head -3  
# Should output JSON – if Permission denied appears, recheck the usermod step
```

Step 4 — Configure Wazuh Agent to Monitor eve.json

NOTE: Run on: AGENT (rakib-virtual-machine)

Edit the agent ossec.conf:

```
sudo nano /var/ossec/etc/ossec.conf
```

Add this block before :

```
<localfile>
  <log_format>json</log_format>
  <location>/var/log/suricata/eve.json</location>
</localfile>
```

NOTE: *log_format json* tells Wazuh to parse each line as JSON. This is required for the Suricata rules to match correctly.

Restart the Wazuh agent:

```
sudo systemctl restart wazuh-agent
sudo systemctl status wazuh-agent
# Expected: Active: active (running)
```

Verify Wazuh is reading the file:

```
sudo grep "eve.json" /var/ossec/logs/ossec.log
# Expected: INFO: (1950): Analyzing file: '/var/log/suricata/eve.json'
```

Step 5 — Add Custom Suricata Rule on the Manager

NOTE: Run on: MANAGER (wazuh-soc)

Wazuh has a built-in Suricata rule file at `/var/ossec/ruleset/rules/0475-suricata_rules.xml`. The base alert rule (86601) is set to level 3 — below most alert thresholds. We override it with a custom rule at level 7 so it appears in the dashboard.

WARNING: Do NOT edit files under `/var/ossec/ruleset/rules/` — these are managed by Wazuh and will be overwritten on updates. Always use `/var/ossec/etc/rules/local_rules.xml` for custom rules.

Edit `local_rules.xml`:

```
sudo nano /var/ossec/etc/rules/local_rules.xml
```

Add this block inside the existing tag (or before):

```
<group name="ids,suricata,">
  <rule id="86701" level="7">
    <if_sid>86601</if_sid>
    <description>Suricata: Alert - $(alert.signature)</description>
    <group>network,ids,suricata,</group>
  </rule>
</group>
```

NOTE: Rule 86701 inherits from 86601 (the base Suricata alert rule) and raises its level to 7. The `alert.signature` field shows the exact rule name that matched, e.g. "GPL ATTACK_RESPONSE id check returned root".

Restart Wazuh Manager:

```
sudo systemctl restart wazuh-manager
```

Verify with `wazuh-logtest`:

```
sudo /var/ossec/bin/wazuh-logtest
# Paste this test JSON:
{"timestamp":"2026-06-24T12:06:14+0600","event_type":"alert",
 "src_ip":"13.35.20.90","dest_ip":"192.168.79.131",
 "alert":{"signature":"GPL ATTACK_RESPONSE id check returned root","severity":2}}
# Expected output:
# id: '86701'
# level: '7'
# Alert to be generated.
```


Step 6 — Create the Auto-Block Script on the Agent

NOTE: Run on: AGENT (rakib-virtual-machine)

This script reads the Suricata alert JSON from stdin (sent by Wazuh execd), extracts the attacker's IP, and blocks it using iptables.

WARNING: Wazuh 4.x active response scripts receive the alert JSON via stdin — NOT as command-line arguments. Scripts that read \$1/\$2/\$3/\$4 will fail with "Cannot read srcip from data". Always use read INPUT at the top of the script.

Create the script:

```
sudo nano /var/ossec/active-response/bin/suricata-drop.sh
```

Paste the complete script:

```
#!/bin/bash
LOG=/var/ossec/logs/active-responses.log
# Read alert JSON from stdin (Wazuh 4.x method)
read -r INPUT
# Extract attacker IP from nested JSON
SRC_IP=$(echo "$INPUT" | grep -oP '"parameters".*?"alert".*?"data".*?"src_ip": "\K[^\"]+' | head -1)
if [ -z "$SRC_IP" ]; then
    echo "$(date) - suricata-drop: Cannot find src_ip" >> $LOG
    exit 1
fi
# Check if already blocked – prevent duplicate iptables entries
if iptables -L INPUT -n | grep -q "$SRC_IP"; then
    echo "$(date) - suricata-drop: $SRC_IP already blocked, skipping" >> $LOG
    exit 0
fi
echo "$(date) - suricata-drop: Blocking IP $SRC_IP" >> $LOG
iptables -I INPUT -s $SRC_IP -j DROP
iptables -I FORWARD -s $SRC_IP -j DROP
exit 0
```

Set correct permissions:

```
sudo chmod 750 /var/ossec/active-response/bin/suricata-drop.sh
sudo chown root:wazuh /var/ossec/active-response/bin/suricata-drop.sh
# Verify
sudo ls -la /var/ossec/active-response/bin/suricata-drop.sh
# Expected: -rwxr-x--- 1 root wazuh ... suricata-drop.sh
```

Step 7 — Configure Active Response on the Manager

NOTE: Run on: *MANAGER* (*wazuh-soc*)

Edit `ossec.conf`:

```
sudo nano /var/ossec/etc/ossec.conf
```

Add these two blocks before :

```
<!-- Command definition – must come before active-response -->
<command>
  <name>suricata-drop</name>
  <executable>suricata-drop.sh</executable>
  <timeout_allowed>yes</timeout_allowed>
</command>

<!-- Active response – triggers when rule 86701 fires -->
<active-response>
  <command>suricata-drop</command>
  <location>local</location>
  <rules_id>86701</rules_id>
  <timeout>600</timeout>
</active-response>
```

Settings explained:

Setting	Value	Meaning
location	local	Run on the agent where the alert originated
rules_id	86701	Only trigger for our Suricata rule
timeout	600	Auto-unblock the IP after 10 minutes

Restart Wazuh Manager:

```
sudo systemctl restart wazuh-manager
sudo systemctl status wazuh-manager
```

Step 8 — Test the Full Pipeline

NOTE: Run on: AGENT (rakib-virtual-machine)

testmynids.org is a website specifically built to test Suricata/Snort detection. When you curl it, it responds with "uid=0(root) gid=0(root) groups=0(root)" — a string that matches Suricata rule 2100498 (GPL ATTACK_RESPONSE id check returned root).

Open two terminals on the agent:

Terminal 1 — watch the active response log:

```
sudo tail -f /var/ossec/logs/active-responses.log
```

Terminal 2 — send the test request:

```
curl http://testmynids.org/uid/index.html
# Expected response: uid=0(root) gid=0(root) groups=0(root)
```

Expected output in Terminal 1 (first curl):

```
Wed Jun 24 12:54:56 2026 - suricata-drop: Blocking IP 13.35.20.97
```

Run curl again — duplicate block should be skipped:

```
curl http://testmynids.org/uid/index.html
# Expected in Terminal 1:
Wed Jun 24 12:55:10 2026 - suricata-drop: 13.35.20.97 already blocked, skipping
```

NOTE: testmynids.org uses CloudFront CDN so different curl requests may hit different IPs (13.35.20.57, .59, .90, .97). Each unique IP is blocked once and skipped on repeat.

Verify the IP is blocked in iptables:

```
sudo iptables -L INPUT | grep 13.35.20
# Expected: DROP all -- server-13-35-20-97.cloudfront.net anywhere
```

Verify alert in Wazuh Dashboard:

- Go to Security Events in the Wazuh Dashboard
- Search for: rule.groups: suricata
- You should see an alert with rule.id: 86701, level: 7
- data.alert.signature will show: GPL ATTACK_RESPONSE id check returned root
- data.src_ip will show the CloudFront IP that was blocked

Troubleshooting

suricata-update conflict during install

Cause: Suricata 8.x bundles suricata-update internally. The old standalone package conflicts during upgrade.

Fix:

```
sudo apt remove suricata-update -y
sudo apt install suricata -y
```

eve.json Permission denied for wazuh user

Cause: /var/log/suricata/ is owned by the suricata group. The wazuh user needs to be added to it.

Fix:

```
sudo usermod -aG suricata wazuh
sudo systemctl restart wazuh-agent
```

Rule 86701 shows level 3 instead of 7

Cause: The built-in rule 86601 is at level 3 and is loading before your custom rule. Make sure you are using rule ID 86701 (not 86601) in local_rules.xml.

Fix:

```
sudo /var/ossec/bin/wazuh-logtest # paste test JSON and check level
```

Same IP being blocked multiple times (duplicate iptables entries)

Cause: Using iptables -I without checking first inserts a new DROP rule on every alert, even if the IP is already blocked. The updated suricata-drop.sh checks first with iptables -L INPUT -n | grep -q before inserting. To clean up existing duplicates, flush the INPUT chain and restart the agent.

Fix:

```
# Clear all existing iptables rules
sudo iptables -F INPUT

# Restart agent so Wazuh re-applies any needed rules
sudo systemctl restart wazuh-agent

# Verify no duplicates after next block
sudo iptables -L INPUT -n | grep 13.35.20
```

"Cannot read srcip from data" in active-responses.log

Cause: The old firewall-drop script is running instead of suricata-drop.sh. Check that the active-response block in ossec.conf uses suricata-drop.

Fix:

```
sudo grep -B5 "86701" /var/ossec/etc/ossec.conf
# Should show: <command>suricata-drop</command>
```

No new entries appear in active-responses.log after curl

Cause: Suricata may not have rules loaded. Check rules_loaded count.

Fix:

```
sudo suricata -T -c /etc/suricata/suricata.yaml -v 2>&1 | grep "rules successfully loaded"
# Expected: 50861 rules successfully loaded
# If 0 rules: run suricata-update first
sudo suricata-update && sudo systemctl restart suricata
```

Quick Reference

Key File Paths

File / Component	Path
Suricata main config	/etc/suricata/suricata.yaml
Suricata alert log (JSON)	/var/log/suricata/eve.json
Suricata fast alert log	/var/log/suricata/fast.log
Suricata rules directory	/var/lib/suricata/rules/
Built-in Suricata Wazuh rules	/var/ossec/ruleset/rules/0475-suricata_rules.xml
Custom Wazuh rules	/var/ossec/etc/rules/local_rules.xml
Auto-block script (on AGENT)	/var/ossec/active-response/bin/suricata-drop.sh
Active response log (on AGENT)	/var/ossec/logs/active-responses.log
Manager alert log	/var/ossec/logs/alerts/alerts.log
Agent ossec.conf	/var/ossec/etc/ossec.conf (on agent)
Manager ossec.conf	/var/ossec/etc/ossec.conf (on manager)

Useful Commands

What it does	Command
Test Suricata config	sudo suricata -T -c /etc/suricata/suricata.yaml -v
Update Suricata rules	sudo suricata-update
Watch live alerts	sudo tail -f /var/log/suricata/eve.json grep alert
Check Wazuh rule match	sudo /var/ossec/bin/wazuh-logtest
Watch auto-block log	sudo tail -f /var/ossec/logs/active-responses.log
View blocked IPs	sudo iptables -L INPUT grep DROP
Manually unblock an IP	sudo iptables -D INPUT -s <IP> -j DROP
Search Suricata alerts	sudo grep '"event_type": "alert"' /var/log/suricata/eve.json tail -5
Restart Suricata	sudo systemctl restart suricata
Restart Wazuh Manager	sudo systemctl restart wazuh-manager
Restart Wazuh Agent	sudo systemctl restart wazuh-agent

Final Setup Checklist

Go through this every time you set this up on a new server:

Task	Done?
Suricata 8.0.5 installed via OISF stable PPA	☐
suricata-update run — 50,000+ rules loaded	☐
Network interface set to correct name (ens33) in suricata.yaml	☐
eve.json output enabled with alert, http, dns, tls, flow types	☐
suricata -T config test passes with no errors	☐
Suricata service running: systemctl status suricata	☐
wazuh user added to suricata group via usermod -aG	☐
eve.json readable by wazuh user: sudo -u wazuh cat /var/log/suricata/eve.json	☐
localfile block added to agent ossec.conf for eve.json	☐
Wazuh agent log shows: Analyzing file: /var/log/suricata/eve.json	☐
Rule 86701 added to local_rules.xml on Manager at level 7	☐
wazuh-logtest confirms level 7 and "Alert to be generated"	☐
suricata-drop.sh created on AGENT with permissions 750, owner root:wazuh	☐
Manager ossec.conf has <command>suricata-drop</command> block BEFORE <active-response>	☐
curl http://testmynids.org/uid/index.html triggers alert	☐
active-responses.log shows: suricata-drop: Blocking IP ...	☐
iptables -L INPUT shows DROP rule for blocked IP	☐
Wazuh Dashboard shows alert under rule.groups: suricata	☐

-- End of Document --